Milestone 3: Model Architecture & End-to-End Pipeline Verification

Regional Dialect Synthesis Pipeline

Standard Hindi → Haryanvi (Bangru)

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Deadline: 19 March 2026

Project Overview

Dual-Architecture Pipeline

Module A (Text-to-Text): Translating Standard Hindi to Haryanvi (Bangru) dialect using fine-tuned LLaMA 3.1

Module B (Text-to-Speech): Synthesizing translated Haryanvi text into natural, dialect-accurate speech using VITS

> **Milestone 3 Goal:** Verify complete end-to-end pipeline with preprocessing, tokenization, forward passes, and output generations working cohesively without integration errors.

Datasets:

- Text: Satyam-Srivastava/RDS (~5,000 parallel Hindi-Haryanvi pairs)
- Audio: ankitdhiman/haryanvi-tts (~4,000 single-speaker pairs)

Dataset Organization & Splits

Data Structure

- **hf_dataset_cache/:** Full downloaded dataset via snapshot_download()
- **haryanvi_vits_data/:** Processed TTS data in LJSpeech format
- **tts_output/:** Model checkpoints and outputs

Train/Val/Test Splits

Module	Split	Size	Proportion
Text (Module A)	Train	4,011	80%
Text (Module A)	Validation	501	10%
Text (Module A)	Test	502	10%
Audio (Module B)	Train	~90% of filtered	90%
Audio (Module B)	Eval	capped at 50	10%

Module A: Text Preprocessing (LLaMA Translation)

Preprocessing Steps

1. **Data Ingestion:** JSON + parallel .txt files → 5,025 pairs
2. **Deduplication:** drop_duplicates() removes 8 repeated pairs → 5,017
3. **Length Filtering:** Keep only 3–25 word sentences
4. **Train/Val/Test Split:** 80/10/10 with random_state=42
5. **Chat Template:** Format as system/user/assistant dialogue
6. **Tokenization:** LLaMA tokenizer with max_length=128, padding, truncation

> Output: `input_ids` and `attention_mask` tensors ready for SFTTrainer

Module B: Audio Preprocessing (VITS TTS)

Preprocessing Pipeline

1. **Deduplication:** Remove repeated transcripts
2. **Word-count Filtering:** Keep only 3–25 word transcripts
3. **Full Download:** snapshot_download() → local files
4. **Resampling:** All audio standardized to 22050 Hz
5. **LJSpeech Format:** filename|transcript|normalized_transcript
6. **Character-Level Input:** use_phonemes=False with basic_cleaners

> **Why character-level?** espeak-ng lacks native Haryanvi support and mispronounces dialect-specific retroflex sounds. Character input preserves Devanagari directly.

Architecture: Module A (Text Translation)

LLaMA 3.1 + QLoRA

Component	Specification
Base Model	meta-llama/Meta-Llama-3.1-8B-Instruct
Quantization	4-bit NormalFloat (NF4) via bitsandbytes
Adaptation	LoRA on q_proj, k_proj, v_proj, o_proj
Loss Function	Causal Cross-Entropy

> Haryanvi lacks millions of parallel sentences. LLaMA 3.1 already understands Hindi — QLoRA steers that knowledge toward Bangru syntax without catastrophic forgetting.

Architecture: Module B (Text-to-Speech)

VITS Acoustic Model

Key Components:

- **Text Encoder:** Maps characters to hidden representations
- **Stochastic Duration Predictor:** Captures Bangru's syllabic compression
- **Posterior Encoder + Flows:** Maps spectrograms to latent space
- **HiFi-GAN Vocoder:** Direct waveform synthesis

Why VITS Over Alternatives?

Model	Duration	Inference	Low-Resource
VITS	✓ Stochastic	✓ Single pass	✓ High
FastSpeech2	! Deterministic	✓ Two-stage	! Robotic
XTTS	! Auto-regressive	! Slow	! Needs large data

Data Flow Architecture

```
Raw Hindi: "मैं आज बाजार जा रहा हूँ।"
        ↓
    [Length Filter + Dedup]
        ↓
    [LLaMA Tokenizer]
        ↓
INPUT: (batch=4, seq≤128)
        ↓
╔═══════════════════════╗
║ MODULE A: LLaMA 3.1   ║
║ + QLoRA               ║
║ Loss: Cross-Entropy   ║
╚═══════════════════════╝
        ↓
Haryanvi: "मैं आज बाजार जा रया सै।"
        ↓
    [basic_cleaners]
        ↓
    [Character Tokenizer]
        ↓
INPUT: (batch=8, text_len)
        ↓
╔═══════════════════════╗
║ MODULE B: VITS        ║
║ Encoder + Duration    ║
║ Loss: L1 + KL + GAN   ║
╚═══════════════════════╝
        ↓
Output: .wav @ 22050 Hz
```

Input-Output Compatibility & Tensor Shapes

Component	Input Format	Shape	Output
LLaMA Tokenizer	UTF-8 string	—	(batch, ≤128)
LLaMA Model	input_ids + attn_mask	(4, ≤128)	Logits (4, seq, vocab)
LoRA Adapters	Hidden states	rank=16, α=32	Delta states
VITS Encoder	Char IDs	(8, text_len)	(8, hidden, text_len)
HiFi-GAN Vocoder	Mel-spectrogram	(8, 80, mel_len)	Waveform (8, 1, audio_len)

> Module A outputs pure UTF-8 Devanagari. Module B's basic_cleaners accepts it directly—clean handoff between modules.

End-to-End Pipeline Verification

Module A: Text Translation (text-to-text-pipe.ipynb)

- **Loss:** Decreased from 0.094 → 0.090 over 3 epochs
- **LoRA:** Adapters merged and saved successfully
- **Status:** ✓ Gradient flow verified

Module B: Text-to-Speech (tts-pipe.ipynb)

- **Data:** Full dataset downloaded + filtered (3–25 words)
- **Training:** VITS Trainer executed for 1 epoch
- **Output:** Checkpoint saved dynamically (best_model.pth preferred)
- **Status:** ✓ GAN training stabilized, .wav generated

> Pipeline executes without dimension mismatches or integration errors. Data flows correctly from raw Hindi text to Haryanvi speech.

Model Outputs & Examples

Translation Results

Input (Hindi)	Output (Haryanvi)	Note
मैं आज बाजार जा रहा हूँ।	मैं आज बाजार जा रया सै।	✓ Correct verb
यह किताब मेरी है, इसे मत छूना।	या किताब मेरी सै, इसे न छूणा।	✓ Excellent
क्या तुम कल स्कूल जाओगे?	क्या तुम कल स्कूल जाओगे।	! Interrogative lost

TTS Output

- **Input:** "मैं आज बाजार जा रया सै।"
- **Output:** test_haryanvi_output.wav @ 22050 Hz
- **Status:** ✓ Successfully generated (low fidelity expected from 1 epoch)

Loss Functions & Evaluation Metrics

Module A: Text Translation

- **Loss:** Causal Language Modeling Cross-Entropy
- **chrF++:** Character n-gram F-score (optimal for morphologically rich dialects)
- **BLEU:** Lexical overlap benchmark

Module B: Text-to-Speech

- **Generator Loss:** L1 Reconstruction + KL Divergence + Adversarial + Feature Matching
- **Discriminator Loss:** Real vs. Fake classification
- **UTMOS:** Reference-free neural MOS estimator (no ground-truth needed)
- **MOS:** Human evaluation (Milestone 4)

> UTMOS used instead of MCD since pipeline generates novel Haryanvi speech without reference recordings.

Team Contributions

Abhishek

- Designed Hindi → Haryanvi translation pipeline
- LLaMA 3.1 + QLoRA implementation
- Chat template formatting & tokenization
- Inference validation on test data
- TTS compatibility verification

Satyam

- Created Milestone 3 report
- Final PPT work
- Worklog & submission management

Sanket

- Dataset consistency verification
- Hugging Face dataset upload
- Pipeline testing & validation
- PPT structure preparation

Fazlur

- Designed TTS architecture (VITS-based)
- Defined audio preprocessing pipeline
- LJSpeech format structuring
- End-to-end TTS pipeline implementation

Conclusion & Next Steps

Milestone 3 Achievement

> ✓ Complete end-to-end pipeline verified without integration errors
> ✓ Data flows correctly from raw Hindi text to Haryanvi audio
> ✓ All component interactions validated

Next Steps: Milestone 4

1. **Full-Scale Training:** Train Module B on all ~4,000 filtered audio samples
2. **Transfer Learning:** Initialize VITS from AI4Bharat vits_rasa_13 Hindi checkpoint, freeze Text Encoder, fine-tune Duration Predictor
3. **Quantitative Evaluation:** Compute BLEU/chrF++ for Module A and UTMOS for Module B
4. **Human Evaluation:** Collect MOS scores for dialect authenticity
5. **Error Analysis:** Investigate interrogative handling and morphological edge cases